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The effects of two different electrical stimulation methods on the pain intensity of the patients who had undergone abdominal surgery with a midline incision: Randomized controlled clinical trial

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Aims-Objectives: The purpose of this study is to determine the effects of transcutaneous electrical nerve stimulation (TENS) and transcutaneous acupoint electrical stimulation (TAES) on pain and analgesic drug consumption in patients who had undergone abdominal surgery.
Background: Evidence for the effects of and TAES on pain and analgesic consumption on patients undergoing abdominal surgery.
Design: This research was conducted as a randomized controlled trial.
Methods: This research sample consisted of 48 patients who underwent abdominal surgery with a midline incision. The patients were randomized into three groups, the first intervention group, which used TENS, the second intervention group, which used TAES (ST25, P6, ST36, Lİ4) and the control group, which did not.
Results: Pain scores and analgesic consumption of both intervention groups were significantly lower than the control group.
Conclusion: The research's findings reveal that the two electrical stimulation methods have similar effects on pain scores and analgesic consumption in patients.

Keywords: acupuncture; nursing; transcutaneous electric nerve stimulation; complementary therapies postoperative; pain

1. Introduction

The vast majority of study subjects experience pain at a moderate to severe level following surgery (Karatas, Eti, Saracoglu, & Gogus, 2015; Wang et al., 1997). Surgical procedures cause tissue trauma, destruction of nerve fibers, release of chemical mediators, and pain with stimulation of pain receptors as a result of the incision (Rakel & Frantz, 2003). Rosen et al. found patients to have the worst pain scores within the postoperative first 48 h (±4 h) in their study on 435 patients who underwent surgery for orthopedics, general surgery and urology-related disorders (Rosen, Bergh, Oden, & Martensson, 2011). Inadequate pain management after surgery is considered to be the most common cause of increased complication rates (Engen et al., 2016).

It is an indisputable fact that postoperative pain control is necessary. An international study group consisting of surgeons and anesthesiologists has evaluated evidence-based

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implementations and recommended strong opioid treatment (IV PCA) + NSAID and acetaminophen use for postoperative pain management after colon resection or laparoscopic colorectal surgery in the Procedure-specific postoperative pain management (PROSPECT) study (Kehlet, 2005). However, the side effects of these drugs such as sedation, respiratory depression, cardiac inhibition, gastrointestinal problems such as nausea and vomiting, itching and hemorrhage lead to an increase in the bed rest period and a delay in postoperative recovery (Hamza, White, Ahmed, & Ghoname, 1999; Sakurai, Suleman, Morioka, Akca, & Sessler, 2003; Wang et al., 2014; Yeh, Chung, Chen, Tsou, & Chen, 2010). Drugs used in the treatment of these side effects may not be able to fully improve the well-being of the patient. The aims of pain treatment are to select multiple treatment options with multiple mechanisms in pain control, avoiding opioids, and using non-pharmacological methods to ensure lower doses of drugs with fewer side effects (Kara et al., 2011).

The use of non-pharmacological methods in addition to pharmacological treatment in providing control of surgical pain may contribute positively to patient outcomes by decreasing the use of pharmacological agents and also the level of pain experienced by the patient. Nurses have an active role in deciding on non-pharmacological interventions and evaluating the effectiveness of the procedures and patient reactions within the framework of a team mentality (Arslan & Celebioglu, 2004; Ozveren, 2011; Sluka & Walsh, 2003). The main non-pharmacological methods include approaches such as transcutaneous electrical nerve stimulation (TENS) and acupuncture (with or without needles). These methods are easy to learn and inexpensive and decrease the need for pharmacological agents when used in conjunction with pharmacological methods (Ahmed, 2010; Hamza et al., 1999; Kara et al., 2011; Stas, Koeniger-Donohue, Julie, & Hawkins, 2012; Yeh, Chung, Chen, & Chen, 2011). Nurses should know and use these methods for pain control.

2. Background

TENS is a non-pharmacological method used in pain management. Acupuncture involves stimulating acupuncture points by needle penetration, placement of electrodes, or electrical stimulation with an acupuncture wand. The advantages of stimulation by placing electrodes are lack of skin trauma, no risk of infection, easy application, inexpensive use, and no side effects (Tokuda, Tabira, Masuda, Nishiwada, & Shomoto, 2014; Yeh et al., 2010). Bjordal et al. reviewed 21 randomized controlled studies and electrical stimulation was found to decrease analgesic consumption for postoperative pain by 35.5% (Bjordal, Johnson, & Ljunggreen, 2003). Tokuda et al. found that electrical stimulation at varying frequency and intensity implemented to the incision circumference in patients who underwent abdominal surgery was effective on pain (Tokuda et al., 2014). Yeh et al. found significant differences in terms of pain and the amount of analgesic used in the group receiving electrical stimulation to the BL40, GB34, HT7, and P6 acupuncture points in their study on patients who had undergone a surgical procedure for a non-traumatic vertebral injury (Yeh et al., 2011). Kerai et al. concluded in their systematic review of TENS that it was effective for postoperative pain, that the stimuli should be strong but at a level that does not bother the patient to provide effective postoperative analgesia, that the placement site of the electrodes should be 2–3 cm lateral to the surgical incision or at the relevant acupuncture points, and that stimuli at varying frequencies were more useful as they activated different opioid receptors (Kerai, Saxena, Taneja, & Sehrawat, 2014). Chen et al. found significant differences in the Patient Controlled Analgesia (PCA) requests and pain scores in their study where TENS was applied to the relevant dermatome of the surgical incision, the ST36 acupuncture point, and the unassociated shoulder area in patients who had undergone major gynecologic surgery (Chen et al., 1998).

We did not come across any study on the effect of TENS and transcutaneous acupoint electrical stimulation (TAES) implementation on pain scores and analgesic consumption in patients with severe pain and in whom a high-level neuroendocrine stress response was seen after abdominal surgery with a midline incision in our review of the relevant literature. We evaluated the effect of these two methods on pain scores and analgesic consumption in a group of patients who had undergone abdominal surgery with a midline incision in this study.

3. Methods

3.1. *Study design*

This study was conducted as a randomized controlled trial. The primary outcome measures were the effects of TENS and TAES on pain and analgesic drug consumption in patients who had undergone abdominal surgery with a midline incision. Secondary outcome measures were the effects of TENS and TAES use on respiratory functions, vital signs, nausea and vomiting status, dizziness, antiemetic drug consumption, and saturation.

3.2. *Study setting*

This study was conducted at the Department of General Surgery of the Gülhane Training and Research Hospital in Ankara, Turkey between May 2015 and November 2016.

3.3. *Participants*

The population consisted of General Surgery patients who underwent planned abdominal surgery with a midline incision between May 2015 and November 2016. The study sample comprised patients who underwent abdominal surgery with a midline incision due to problems related to colorectal cancer and stomach cancer. The principal investigator evaluated patients awaiting abdominal surgery with a midline incision in the hospital room the day before the surgery to ensure that the patients qualified for the study in accordance with the eligibility criteria.

3.3.1. *Inclusion criteria*

Patients who had an American Society of Anesthesiologists (ASA) score of I–II, were aged over 18 years, who could read and write Turkish, who were scheduled to undergo elective abdominal surgery with a midline incision for a diagnosis of stomach or colorectal cancer, and without any impairment of vision, hearing or speech were included in the study.

3.3.2. *Exclusion criteria*

Patients who had a pacemaker, whose skin integrity around the incision was degraded, had a cognitive disorder, had a history of chronic pain, or were suffering from neurological, renal, cardiac or pulmonary disorders that could affect the test results, and patients with an opioid addiction, those who had previously undergone electrical stimulation treatment, morbidly obese subjects and patients using psychoactive drugs were excluded.

3.3.3. *Randomization*

To determine the groups, randomization was performed with a web-based randomization system with the help of a computer by researchers. A block randomization list was obtained for 3 groups.

TENS was implemented to the incision circumference of the patients in the first group and to the acupuncture points of the patients in the second group with no treatment being implemented to the control group.

3.4. *The interventions*

The standard anesthesia protocol was implemented to all patients suitable for the study. All of the patients included in the study were administered 2 mg midazolam IV 5 min before the operation as premedication, after the peripheral vein was accessed. Once the ECG, noninvasive blood pressure, and pulse oximeter monitorization was started, preoxygenization was provided and anesthesia induction conducted intravenously with propofol 2–2.5 mg/kg, rocuronium bromide 0.6 mg/kg, and fentanyl 2 mcg/kg. This was followed by oral tracheal intubation 90 s after 100% O_2 at 4 l/min was started. After endotracheal intubation, the inhalation agent desflurane was set at 6% concentration. All patients were administered 0.2 mg/kg rocuronium bromide IV at 30-minute intervals and IV remifentanil was adjusted to 0.25 mcg/kg/min. All patients were administered intraoperative ranitidine HCl 20 mg and metoclopramide 10 mg. After the last skin suture was placed, the inhalation agent and remifentanil infusion were terminated. Once spontaneous breathing started, the muscle relaxant agent was antagonized (neostigmin 0.05 mg/kg, atropine 0.015 mg/kg) and the patient was extubated when spontaneous breathing and adequate muscle strength were present.

A standard analgesia protocol for postoperative pain control was used in all patients included in the study. The PCA infusion was administered to the patients for 48 h. The patients were provided a total of 500 mg Tramadol HCl PCA infusion intravenously (IV) at a concentration of 5 mg/ml, an infusion dose of 5 mg/h, a bolus dose of 20 mg, and a 150 mg per 4 h limit with a 30 min lock-in time. Dexketoprofen Trometamol vial IV and/or 50 mg Pethidine HCl IM were used as rescue analgesic drugs.

A standard antiemetic protocol was used for all patients after surgery. If they had nausea, 4 mg/2 ml ondansetron was administered intravenously.

First Intervention Group: The PCA infusion was started right after the surgery. Four electrodes were placed 2–3 cm lateral to the incision of the patients at the 30th minute and 2, 18, 22, 42, and 46th hours after the surgery and electrical stimulation was implemented at varying frequencies of 2–100 Hz for 30 min, at a maximum current intensity of 12 mA that would not bother the patient or create muscle contractions, with a pulse duration of 0.25 min. (Figure 1). The interventions were planned during the daytime to avoid disturbing the circadian rhythm of the patients.

Second Intervention Group: The PCA infusion was started right after the surgery. Four electrodes were placed at the ST25, P6, ST36, and LI4 acupuncture points of the patients at the 30th minute and 2, 18, 22, 42, and 46th hours after the surgery and electrical stimulation was implemented at varying frequencies of 2–100 Hz for 30 min, at a maximum current intensity of 12 mA that would not bother the patient or create muscle contractions, with a pulse duration of 0.25 min. (Figure 2). The interventions were planned during the daytime to avoid disturbing the circadian rhythm of the patients.

Control Group: The PCA infusion was started right after the surgery. No intervention was performed to the patients in the control group. The data of the patients were recorded at the postoperative 30th minute and 2, 18, 22, 42, and 46th hours. The flow chart of the study is shown in Figure 3.

The TENS device used in this study was a dual channel nerve stimulator. This device had seven preset pain programs and three special programs. The program and the amplitude could be set individually for each channel. The electrodes of the device were 4 × 4 cm in size with

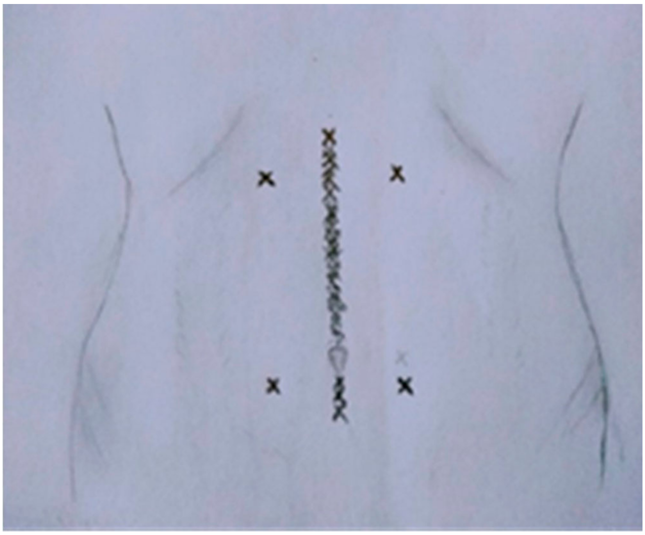

Figure 1. TENS implementation to the incision periphery.

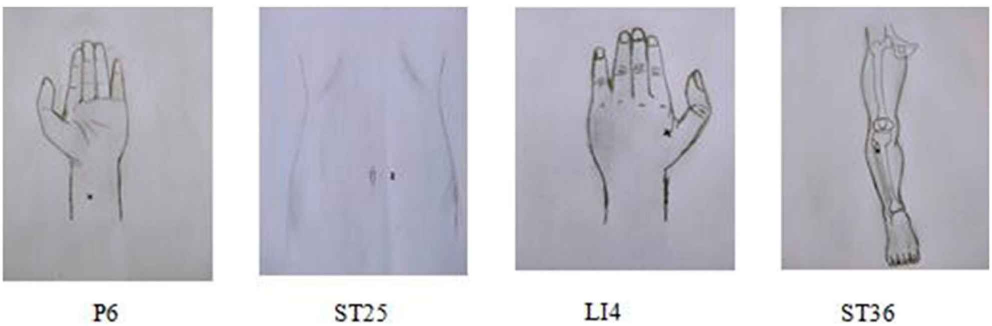

Figure 2. TAES implementation to the acupuncture points.

four pieces in total. The electrodes were placed 2–3 cm lateral to the incision area and at the relevant acupuncture points. The current intensity was manually set according to the patient's tolerance. All interventions are implemented by the responsible investigator. The responsible investigator has been consulted on the application of electrical stimulation and determining acupuncture points. With the use of 4×4 size electrodes, it is not possible to miss the acupuncture points.

3.5. *Measurements*

3.5.1. *Baseline measurements*

Preoperative measurements: The patients' age, gender, height, weight, diagnosis, history of a chronic sleep disorder and ASA class data were recorded. Pulmonary functions (Peak Expiratory Flow: PEF, Forced Vital Capacity: FVC), vital signs, respiratory rate, saturation and pain level data were also recorded.

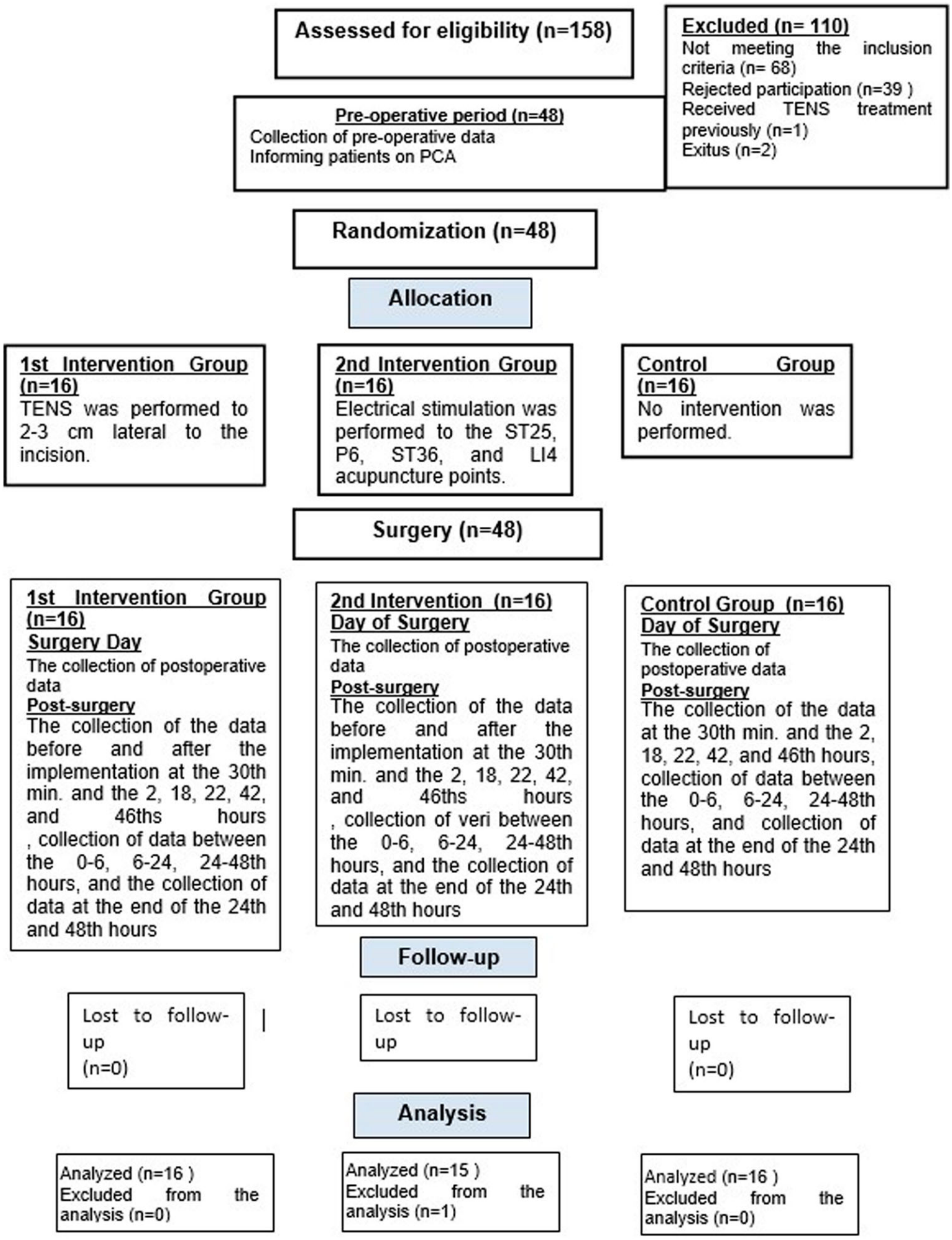

Figure 3. Study flow diagram.

3.5.2. *Outcome measurements*

3.5.2.1. *Intraoperative measurement*. All patients received general anesthesia with the same protocol. The patients were observed during the surgical procedure and data such as incision length, duration of anesthesia, duration of surgery, and number of drains were recorded.

3.5.2.2. *Postoperative measurements*. The medical findings of all patients after the surgical procedure were recorded.

3.5.2.3. *Pain*. VAS with a range of 1–10 points was used while the pain levels of all patients were evaluated. Pain levels of all patients were evaluated at rest.

3.5.2.4. *Analgesic drug consumption*. All patients were administered IV Contramal infusion with PCA for the first 48 h. Dexketoprofen Trametamol via IV and/or 50 mg. Pethidine HCl IM was used as the rescue analgesic.

3.5.2.5. *Nausea severity*. VAS with a range of 1–10 points was used while evaluating the nausea severity of all patients.

3.5.2.6. *Vomiting*. The number of times the patients vomited was recorded.

3.5.2.7. *Antiemetic drug consumption*. The antiemetic drug amount administered to the patients was recorded in mg.

3.5.2.8. *Pulmonary function tests*. The pulmonary functions of the patients were evaluated with a portable spirometer. PEF and FVC values were recorded. The measurement of respiratory functions was performed with the patient wrapping the disposable mouthpiece well with the lips during the measurement. The patient was asked to breathe deeply and exhale suddenly and quickly. This process was repeated three times and the best performance was taken as the base value.

3.6. *Sample size*

The software package G-power 3.1.3. was used to conduct a prior power analysis to calculate the number of participants required. Sixteen participants were required to detect an effect size of 1.4 at 80% power. The alpha level used to define significance was 0.05.

3.7. *Blinding*

Because the patients and data collector were aware they were implementation electrical stimulation or not, there was no blinding to the study.

3.8. *Data collection*

Data collection was started on the day before the patients underwent surgery and finished on the second day after surgery.

3.8.1. *Preoperative data collection*

The characteristics of the patients and ASA classifications, vital signs and respiratory functions were evaluated one day prior to the surgery.

3.8.2. *Postoperative data collection*

The medical information of the patients was recorded within the initial 48 h after the surgery. Vital signs, respiratory functions, pain levels, nausea severity, vomiting status, dizziness status, analgesic and antiemetic drug consumption of all patients included in the study were recorded. Vital signs and pain levels of the patients in the control group were evaluated at the postoperative 30th minutes and 2nd, 18th, 22th, 42th, and 46th hours. Vital findings and pain levels of the patients who were included in the intervention groups before and 30 min after the implementation were evaluated. All evaluations of the patients were performed during bed rest. Nausea levels, vomiting status and dizziness status of the patients were evaluated at the periods of 0–6, 6–24, and 24–48 h. Analgesic and antiemetic drug consumption of the patients was evaluated at the periods of 0–24 and 24–48 h. Respiratory function evaluations of the patients were performed at the 24th and 48th hours.

3.9. *Data analysis*

The data obtained from this study were evaluated by using the Statistical Package for the Social Sciences (SPSS,Inc., Chicago, IL, USA) 22.0 software program.

After the data were transferred to the SPSS database, error controls were performed. Their compliance with a normal distribution was evaluated with Kolmogorov–Smirnov test. The data determined were mean ± standard deviation ($X \pm$ SD), median and minimum–maximum (min–max) for the variables determined by the measurement and number (*n*) and percent (%) for the variables determined by counting. Since the variables determined by counting did not comply with a normal distribution, the difference between the three groups was evaluated by the Chi-square test. The one-way ANOVA test was used to investigate the difference between the three groups regarding the measured values for values complying with normal distribution and the Kruskal–Wallis test was used for the values that did not comply with a normal distribution. The Bonferroni correction was implemented to determine the source of the difference in group comparisons. The Mann–Whitney *U* test was used to investigate the difference between two groups of variables. A p-value <.05 was accepted as statistically significant. The paired sample *t*-Test was used for values complying with a normal distribution and the Wilcoxon Test was used for values not complying with a normal distribution when comparing the values of patients in the same group before and after the implementation.

3.10. *Ethical consideration*

Written permission and approval was obtained from the Gülhane Military Medical Academy's Head of Clinical Studies Ethics Committee before the study started. Physicians and nurses working at the Gülhane Military Medical Academy, General Surgery Clinic, PostOperative Care Unit and Post-Anesthesia Care Unit within the Anesthesiology and Reanimation Clinic were informed of the study and verbal approval was obtained. The patients were informed about the research. Informed consent was received from patients.

4. Results

The study was performed with 48 patients who had undergone abdominal surgery with a midline incision for a diagnosis of stomach and colorectal cancer, complied with study inclusion criteria, and accepted to participate in the study. The patients were randomly divided into three groups.

There were 16 patients in the first intervention group, 16 in the second intervention group, and 16 in the control group. One patient was excluded from the analysis due to non-protocol drug use. The distribution of the groups according to the introductory characteristics of the patients and the factors constituting risk in terms of pain are included in Table 1. Although not specified in the table none of the patients had pain in the preoperative period. The difference between the groups in terms of the values given in the table was not statistically significant ($p > .05$) and the groups were similar. There was no statistically significant difference between groups according to preoperative fever, pulse, blood pressure, respiration rate, PEF, FVC and saturation values

Table 1. Distribution of patient groups according to risk factors for pain ($n = 47$).

Characteristics	1.Intervention Group $n = 16$	2.Intervention Group $n = 15$	Control Group $n = 16$	p			
Age							
$\overline{X} \pm$ SD	60.88 ± 14.84	61.87 ± 16.80	56.31 ± 9.39	.512*			
Med (Min–Max)	63(22–86)	69(23–80)	57.50(40–70)				
BMI (kg/m^2)							
$\overline{X} \pm$ SD	26.44 ± 4.05	26.30 ± 4.16	25.90 ± 4.29	.931*			
Med (Min–Max)	24.96 (20.57–34.61)	25.00 (20.28–36.06)	26.21 (18.82–34.19)				
Anesthesia time							
$\overline{X} \pm$ SD	263.38 ± 143.81	266 ± 95.34	313.75 ± 82.49	.362*			
Med (Min–Max)	217.50 (110–580)	230 (165–505)	300 (165–500)				
Surgery time							
$\overline{X} \pm$ SD	244.88 ± 138.63	247.33 ± 93.88	287.19 ± 83.21	.472*			
Med (Min–Max)	205(95–545)	215(150–480)	270(150–490)				
PACU time							
$\overline{X} \pm$ SD	64.38 ± 44.38	65 ± 24.71	80.63 ± 34.10	.181**			
Med (Min–Max)	60(25–220)	60(45–120)	82.50(30–150)				
	n	%	n	%	n	%	P
Gender							
Female	5	31.25	5	33.3	5	31.25	.990***
Male	11	68.75	10	66.6	11	68.75	
Cancer related surgery							
Stomach cancer	7	43.8	3	20	5	31.3	.365***
Colorectal cancer	9	56.3	12	80	11	68.8	
ASA							
I	7	43.8	4	26.7	6	37.5	.607***
II	9	56.3	11	73.3	10	62.5	
Number of drain							
One	9	56.3	12	80	9	56.3	.295**
Two	7	43.8	3	20	7	43.8	

$\overline{X}$: Mean; SD: Standard Deviation; Med: Median; Min: Minimum; Max: Maximum.
*One-Way Anova;**Kruskal–Wallis Test; ***Chi-Square Test.

($p > .05$) and the groups were similar. Before and after implementation pain severity distribution of the patients in the intervention groups are shown in Table 2. The before and after implementation difference in terms of pain scores was found to be statistically significant ($p < .05$).

The distribution of the patients according to the after implementation pain value are shown in Table 3. The difference between the intervention groups and the control group according to the pain scores was statistically significant ($p < .05$). No significant difference was found between the intervention groups in terms of pain severity ($p > .05$).

The difference between the groups in terms of PEF, FVC and saturation values at the 24th and 48th hours were not statistically significant. The difference between the groups in terms of nausea severity between the 0–6, 6–24 and 24–48th hour periods was not statistically significant. The difference between the groups in terms of the presence of vomiting and dizziness between the 0–6, 6–24 and 24–48th hour periods was not statistically significant.

The distribution of the groups according to the analgesic drug requirement at the postoperative 0–24 and 24–48 h is presented in Table 4. The difference between the groups in terms of consumption of Tramadol HCl administered by PCA was statistically significant ($p < .05$). Tramadol HCl drug consumption of the patients receiving PCA was lower in the first intervention group (TENS implementation to the incision circumference) than the control group and this difference was statistically significant ($p < .05$). The amount of Tramadol HCl consumption was also lower in the second intervention group (electrical stimulation to acupuncture points) than the control group and this difference was statistically significant ($p < .05$). No significant difference was found between the first and second intervention groups in terms of Tramadol HCl consumption ($p > .05$). The difference between the groups in terms of antiemetic drug consumption at the

Table 2. Before and after implementation pain values in interventions groups.

	1. Intervention group			2. Intervention group		
Pain severity	Before implementation	After implementation	p	Before implementation	After implementation	p
2. Hour						
$\overline{X} \pm SD$	6.63 ± 1.89	4.75 ± 1.57	0.001*	5.47 ± 1.64	3.67 ± 1.44	.001*
Med (Min–Max)	7.50(3–9)	5(2–7)		5(3–9)	3(2–7)	
18. Hour						
$\overline{X} \pm SD$	4.25 ± 1.88	2.19 ± 1.27	0.000**	3.93 ± 1.33	2.67 ± 1.23	.001*
Med (Min–Max)	4(1–8)	2(0–5)		4(2–7)	3(1–6)	
22. Hour						
$\overline{X} \pm SD$	4.31 ± 1.25	2.19 ± 1.22	0.000*	4.60 ± 1.29	3 ± 1.30	.000*
Med (Min–Max)	4(2–6)	2(0–5)		5(2–7)	3(1–6)	
42. Hour						
$\overline{X} \pm SD$	3.63 ± 1.25	1.56 ± 1.15	0.000*	4 ± 1.36	2.27 ± 1.28	.001*
Med (Min–Max)	4(2–6)	1(0–4)		4(1–6)	2(0–5)	
46. Hour						
$\overline{X} \pm SD$	3.38 ± 1.25	1.75 ± 0.85	0.000*	3.67 ± 1.29	2.33 ± 0.81	.002*
Med (Min–Max)	3(0–5)	2(0–3)		4(1–7)	2(1–4)	

$\overline{X}$: Mean; SD: Standard Deviation; Med: Median; Min: Minimum; Max: Maximum.
*Wilcoxon Test, **T Test.

Table 3. After implementation pain values in groups.

Pain Severity	1.Intervention Group	2.Intervention Group	Control Group	*p*
2. Hour				
$\overline{X} \pm SD$	4.75 ± 1.57	3.67 ± 1.44	6.38 ± 2.09	.001*
Med (Min–Max)	5(2–7)	3(2–7)	6(3–9)	
18. Hour				
$\overline{X} \pm SD$	2.19 ± 1.27	2.67 ± 1.23	5.81 ± 2.31	.000*
Med (Min–Max)	2(0–5)	3(1–6)	6(2–9)	
22. Hour				
$\overline{X} \pm SD$	2.19 ± 1.22	3 ± 1.30	4.88 ± 1.78	.000*
Med (Min–Max)	2(0–5)	3(1–6)	4.50(3–8)	
42. Hour				
$\overline{X} \pm SD$	1.56 ± 1.15	2.27 ± 1.28	4.19 ± 1.60	.000*
Med (Min–Max)	1(0–4)	2(0–5)	4(2–8)	
46. Hour				
$\overline{X} \pm SD$	1.75 ± 0.85	2.33 ± 0.81	4 ± 1.46	.000*
Med (Min–Max)	2(0–3)	2(1–4)	3(2–7)	

$\overline{X}$: Mean; SD: Standard Deviation; Med: Median; Min: Minimum; Max: Maximum.
*Kruskal–Wallis Test.

Table 4. Distribution of patient groups according to analgesic drugs consumption.

Analgesic drugs consumption	1.Intervention Group	2.Intervention Group	Control Group	*p*
Tramadol HCl (0–24 h)				
$\overline{X} \pm SD$	257.68 ± 65.55	228.40 ± 87.89	357.81 ± 123.70	.001*
Med (Min–Max)	262.25 (135–375)	195 (115–406)	337.25 (197–682)	
Deksketoprofen trametamol (0–24 h)				
$\overline{X} \pm SD$	46.87 ± 46.43	53.33 ± 35.18	71.87 ± 40.69	.220**
Med (Min–Max)	50(0–150)	50(0–100)	50(0–150)	
Pethidine HCl (0–24 h)				
$\overline{X} \pm SD$	33.75 ± 56.55	20.66 ± 27.89	33.12 ± 42.69	.785**
Med (Min–Max)	0(0–200)	0(0–80)	15(0–130)	
Tramadol HCl (24–48 h)				
$\overline{X} \pm SD$	195.50 ± 71.74	164.46 ± 61.31	265.65 ± 96.71	.003*
Med (Min–Max)	177 (100–325)	151 (79–285)	232.50 (124–434.50)	
Deksketoprofen trametamol (24–48 h)				
$\overline{X} \pm SD$	31.25 ± 35.93	33.33 ± 40.82	53.12 ± 59.07	.598**
Med (Min–Max)	25(0–100)	0(0–100)	25(0–150)	
Pethidine HCl (24–48 h)				
$\overline{X} \pm SD$	6.31 ± 17.05	10 ± 20.70	6.25 ± 17.07	.847**
Med (Min–Max)	0(0–50)	0(0–50)	0(0–50)	

$\overline{X}$: Mean; SD: Standard Deviation; Med: Median; Min: Minimum; Max: Maximum.
*One-way Anova; ** Kruskal–Wallis Test.

0–24th and 24–48th hours after surgery was not statistically significant. In addition, body temperature, pulse rate, blood pressure, respiratory rate and saturation values of the patients before and after the implementation were similar in all patients. The difference between the groups in terms of the presence of vomiting and dizziness between the 0–6, 6–24 and 24–48th hour periods was not statistically significant.

5. Discussion

Postoperative pain is a problem experienced by patients due to tissue damage caused by the surgical procedure. It is also something that needs to be taken under control by health care providers. Uncontrolled pain affects all systems negatively and increases the risk of developing complications (Hong & Lee, 2014). Effective pain management improves the life quality of the patient and decreases the cost of health care by shortening the hospitalization duration.

The use of non-pharmacological methods in addition to pharmacological methods in postoperative pain management is important as they are non-invasive and inexpensive with no side effects and can be implemented easily and may reduce the need for drugs in addition to drug adverse effects (Ozdag, Mollahaliloglu, Oztas, & Guzeldemirci, 2015). Nurses working in surgical clinics play a key role in postoperative pain management. The most common problem of patients at the early postoperative stage following abdominal surgery has been reported to be pain in the study of İzveren and Dal the same study found the most common nursing practice performed by the nurses for pain to be drug treatment according to physician orders in the same study (Izveren & Dal, 2011). The nurses were found to make very little use of non-pharmacological methods in the study of Celik (Celik, 2013). Evaluation of the use and effects of pharmacological and non-pharmacological methods will affect the professional development of the nursing profession positively (Sluka & Walsh, 2003). Nurses knowing the expectations of the patients, cooperating with them, and knowing and implementing non-pharmacological methods that are among evidence-based implementations for postoperative pain management will lead to better patient outcomes (Hong & Lee, 2014; Kadıoglu, Turker, Gurbet, Demirci, & Hulagu, 2013).

Postoperative pain is a type of nociceptive pain caused by tissue trauma due to the surgical incision (Kehlet, 2005; Ward, 2014). Rosen et al. found in their study that the patients experienced the most severe pain within the initial 48 h after the surgery (Rosen et al., 2011). Seventy-six percent of the patients who underwent abdominal surgery expressed that pain decreased their comfort within the postoperative initial 6–8 h and 82% within 24 h in the study of Robleda et al. (Robleda, Roche-Campo, Sanchez, Gich, & Banos, 2015). Based on these results, patient complaint of pain indicates the need for the effective use of pharmacological and non-pharmacological methods for postoperative pain management. The use of non-pharmacological methods has become widespread globally in recent years. Parallel to the developments in the health care field worldwide, progress and changes have also been seen in this area in our country. The effect of the non-pharmacological methods of TENS and TAES on postoperative pain was evaluated in this study.

5.1. *Discussion of findings regarding pain severity*

Kerai et al. and Bjordal et al. reported in their systematic review that TENS was effective for postoperative pain control and that the electrodes should be placed to the incision circumference or acupuncture points related to the pain while administering TENS. The use of the highest current intensity that the patient could tolerate at varying frequencies was more effective. TENS implementation at various frequencies could be more effective because electrical stimulation at low and high frequencies activates different opioid receptors (Bjordal et al., 2003; Kerai et al., 2014). Electrical stimulation was performed at the maximum level that patients could tolerate at varying frequencies using four electrodes in our study.

Pain severity decreased after TENS implementation to the incision circumference of the first intervention group at the 2nd, 18th, 22nd, 42nd, and 46th hours compared to the pre-implementation state in our study. TENS implementation was found to be effective in decreasing the pain in the intervention group when compared with the control group as well. Tokuda et al. concluded that the implementation of TENS to the incision circumference after abdominal surgery decreased the pain severity of the patient (Tokuda et al., 2014). Hamza et al. found the implementation of TENS at varying frequencies to be more effective in their study (Hamza et al., 1999). The pain severity scores were found to be lower than the control group in a study where the effect of TENS implementation to the incision circumference following colon surgery on pain was evaluated (Bjersa & Andersson, 2014). Kara et al. and Unterrainer et al. reported that TENS implementation with electrodes placed at the incision circumference of patients who underwent spinal surgery was effective in pain control (Kara et al., 2011; Unterrainer et al., 2010). Similar results were obtained in the studies where the effectiveness of TENS implementation after inguinal hernia surgery on pain was evaluated (DeSantana et al., 2008). The implementation of TENS to the incision circumference of women who underwent cesarean section was reported to decrease the pain in another study where its effect on pain was evaluated (Kayman-Kose et al., 2014). TENS implementation was also found to be effective in decreasing pain in studies conducted on different patient groups in the literature (da Silva, Liebano, Rodrigues, Abla, & Ferreira, 2015; Silva, de Melo, de Oliveira, Crema, & Fernandes, 2012). In contrast to these results, it was concluded that TENS implementation was not effective in a study where its effect on pain after pancreas resection surgery was evaluated (Bjersa & Andersson, 2014). Similarly, TENS was found to be ineffective for pain in a study where its effect after total knee prosthesis surgery was evaluated (Breit & Van der Wall, 2004). TENS implementation on patients who underwent thoracotomy or sternotomy was found to be effective for postoperative pain in other studies (Engen et al., 2016; Feng et al., 2016; Sbruzzi, Silveira, Silva, Coronel, & Plentz, 2012). The presence of different results may be due to the differences in frequency, current intensity, implementation periods and durations of TENS use and it is possible that the scales for pain evaluation could not be accurately interpreted by patients (Bjersa & Andersson, 2014; Kerai et al., 2014; Tokuda et al., 2014). In conclusion, there are studies both supporting and not supporting the effect of TENS implementation. TENS implementation was effective in decreasing the pain in the group of patients who underwent abdominal surgery with a midline incision in our study. Based on these results, TENS implementation being used by the nurses in addition to the pharmacologic methods could be effective in decreasing the pain severity of the patients.

The pain severity of the patients decreased with electrical stimulation to the acupuncture points at the 2nd, 18th, 22nd, 42nd, and 46th hours in the second intervention group in our study compared to the pre-implementation state. When compared with the control group, electrical stimulation to the acupuncture points was found to be effective in reducing the pain of the patients. Electrical stimulation of acupuncture points with electrodes was found to have more positive effects on pain in a study comparing electrical stimulation of acupuncture points with needles or electrodes following gynecologic surgery (Stas et al., 2012). It was concluded in a systematic review on the effect of acupuncture on pain that the method has positive effects (Wu et al., 2016). Similarly, implementing electrical stimulation to acupuncture points in various patient groups was found to have positive effects on pain (Chen et al., 2015; Feng et al., 2016; Francis & Johnson, 2011; Lan, Ma, Xue, Wang, & Ma, 2012; Wang et al., 2014; Yeh et al., 2010; Yeh et al., 2011). The superior points of electrical stimulation with electrodes to acupuncture with a needle are the non-invasive procedure, no risk of infection, no pain, easier detection of the relevant points due to the size of the electrodes and the possibility of administration by the health care staff with minimal effort (Chen et al., 2015; Yeh et al., 2010). Acupuncture points were stimulated with electrodes in our study. The advantages we found were the lack of pain during the procedure, no risk of infections as the procedure was non-invasive, and the ease of implementation, similar to the advantages mentioned in the literature regarding the method. All the studies concluded that the method is safe and effective (Chen et al., 2015; Feng et al., 2016; Wang et al., 2014; Yeh et al., 2010; Yeh et al., 2011). Considering these results, the use of electrical stimulation methods by the nurses who have important responsibilities in the evaluation of the patient's pain status and pain management should improve patient outcomes.

No difference was found in terms of pain severity between the intervention groups in our study ($p > .05$). We did not come across any study where the effect of the two methods on pain in patients who have undergone abdominal surgery with a midline incision was evaluated in the literature. Our findings indicate that both methods are similarly effective on pain. They could therefore be used instead of each other when a medical condition prevents one from being performed. We believe the most appropriate method for the patient should be selected and used when skin integrity at the relevant area is disrupted, the incision area is too sensitive for the procedure, it is not possible to remove the abdominal girdle, and the patient does not want any procedure on the incision area.

5.2. *Discussion of findings related to analgesic consumption*

The consumption amount of Tramadol HCl implemented with PCA in both the first intervention group (TENS implementation to the incision circumference) and the second intervention group (implementation of electrical stimulation to the acupuncture points) was significantly lower than the control group in our study. Similar results were found within the framework of the standard analgesic drug protocol determined in studies in which both interventions were evaluated in the literature (Chen et al., 1998; Chen et al., 2015; DeSantana et al., 2008; Kara et al., 2011; Kayman-Kose et al., 2014; Kerai et al., 2014; Sbruzzi et al., 2012; Unterrainer et al., 2010; Wang et al., 2014; Yeh et al., 2010; Yeh et al., 2011). In contrast, it was concluded that TENS implementation did not have an effect on pain severity and analgesic drug consumption in two different studies in patients who underwent pancreatic resection and total knee arthroplasty surgery (Bjersa & Andersson, 2014; Breit & Van der Wall, 2004). Considering our study results, both electrical stimulation methods decreased the amount of analgesic consumption in patients who underwent abdominal surgery with a midline incision and these electrical stimulation methods can therefore be used to reduce postoperative pain depending on the patient’s condition.

Clinical implications

There was no difference between the two methods implemented to the incision circumference and acupuncture points in terms of pain intensity and analgesic drug use in patients who underwent abdominal surgery with a midline incision. We therefore recommend new studies that compare these two methods in a larger sample and on a homogeneous group of patients.

Limitations

- The limitations of this study were that it was a thesis study and the sample is limited to 47 patients due to the time restriction. It is recommended to conduct new studies with larger samples.
- Because the surgeons prefer the laparoscopic approach, the number of patients with open surgery is limited.
- Patients do not accept the electrical stimulation decreased participation in the study.

6. Conclusion

It was found that two different electrical stimulation methods (electric stimulation to the incision circumference and acupuncture points) were effective on decreasing the severity of pain and analgesic drug consumption in patients who underwent abdominal surgery with a midline incision when compared with the control group and there was no significant difference between the two methods. It is important to use nonpharmacologic methods to reduce pharmacological drug using and side effects in postoperative pain control. To know and implementation nonpharmacologic methods mentioned in the literature for nurses will provide positive contributions to patient outcomes.

Conflict of interest

No conflict of interest has been declared by the authors.

Impact statement

It has been determined that many studies evaluating the effect of TENS and TAES on postoperative pain levels and analgesic consumption in different groups of patients underwent surgery have positive effects. In some studies it was determined that there was no effect.

A randomized controlled trial comparing TENS and TAES was not found in the literature on patients who underwent abdominal surgery with a midline incision.

In this study, it was determined that two different electrical stimulation were positive and similar effects on pain and analgesic consumption in the group of patients who underwent abdominal surgery with a midline incision.

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